Snehal **Raj**

SORBONNE UNIVERSITY · UNIVERSITY OF OXFORD · IIT KANPUR

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Research Summary

My research explores quantum and quantum-inspired algorithms at the intersection of theory and practice. I develop theoretical frameworks for quantum machine learning models and validate them empirically on near-term hardware and classical simulations. I also design classical methods inspired by quantum structures, bridging quantum computing and practical machine learning.

Education

Sorbonne University
Paris, France

PHD IN QUANTUM MACHINE LEARNING — *2024 - Present*

- Supervised by Prof. Elham Kashefi, jointly with QC Ware Corp.
- Research on quantum neural networks, equivariant architectures, fermionic linear optics, and quantum-inspired ML.

University of Oxford
Oxford, United Kingdom

MSC IN ADVANCED COMPUTER SCIENCE — *2021 - 2022*

- Thesis advisor: Prof. Aleks Kissinger.
- Thesis on enhancing quantum circuit simulation via ZX-calculus and tensor network methods [thesis].

Indian Institute of Technology, Kanpur (IIT-K)
India

B.TECH IN COMPUTER SCIENCE AND ENGINEERING — *2017 - 2021*

- GPA: 9.52/10. Undergraduate thesis on polynomial methods in quantum query complexity with Prof. Rajat Mittal.

Experience

QC Ware Corp
Paris, France (prev. Remote)

ASSOCIATE STAFF SCIENTIST (PREV. RESEARCH CONTRACTOR) — *Aug. 2020 - Present*

- Conducted quantum algorithms research applied to quantitative finance in collaboration with major financial institutions including JPMC, Itau Unibanco, Wells Fargo, and UBS.
- Core developer of QC Ware Corp's Quantum Neural Network library, built from scratch in JAX with production-grade integration for PyTorch, TensorFlow, and Scikit-learn. Distributed on AWS Marketplace.
- Led proof-of-concept quantum computing projects for industrial clients including Aisin Corp, BMW, and Airbus.

Adobe Research
Bangalore, India (Remote)

RESEARCH INTERN, UNDER DR. NIYATI CHHAYA — *Apr. 2020 - Aug. 2020*

- Deployed an AutoML pipeline for Adobe's Document Cloud to automatically distill knowledge from large language models like Bert/GPT-2 and deploy models for downstream tasks like summarization.
- The work was accepted at EMNLP 2021 and led to an invention disclosure according to Adobe's policies.

Publications

[1] **The Hamming Weight Landscape of Fermionic Linear Optics: Snehal Raj***, Elham Kashefi — **In Preparation 2026**

[2] **QuIC: Quantum-Inspired Compound Adapters for Parameter Efficient Fine-Tuning: Snehal Raj***, Brian Coyle — **In Submission 2026**

[3] **Quantum Graph Neural Networks with Permutation-Equivariant Message Passing: Snehal Raj***, Brian Coyle, Léo Monbroussou, André J. Ferreira-Martins, Renato M. S. Farias, Elham Kashefi — **In Submission 2026**

[4] **Training Quantum Circuits with Directional Derivatives:** Brian Coyle, **Snehal Raj***, El Amine Cherrat — **In Submission 2026**

[5] **Bayesian Quantum Orthogonal Neural Networks for Anomaly Detection:** Natansh Mathur, Brian Coyle, Nishant Jain, **Snehal Raj***, Akshat Tandon, Jasper Simon Krauser, Rainer Stoessel — **IEEE QW 2025**

[6] **Training-efficient density quantum machine learning:** Brian Coyle, **Snehal Raj***, Natansh Mathur, El Amine Cherrat, Nishant Jain, Skander Kazdaghli, Iordanis Kerenidis — **NPJ QI 2025**

[7] **Quantum Deep Hedging:** El Amine Cherrat*, **Snehal Raj***, Iordanis Kerenidis*, Abhishek Shekhar, Ben Wood, Jon Dee, Shouvanik Chakrabarti, Richard Chen, Dylan Herman, Shaohan Hu, Pierre Minssen, Ruslan Shaydulin, Yue Sun, Romina Yalovetzky, Marco Pistoia. **[forbes press release]**. — **Quantum 2023**

[8] **AUTOSUMM: Automatic Model Creation for Text Summarization:** Sharmila Reddy Nangi, Atharv Tyagi*, Jay Mundra*, Sagnik Mukherjee*, **Snehal Raj***, Niyati Chhaya, and Aparna Garimella. (* *equal contribution*) — **EMNLP 2021**

** denotes equal first authorship or equal contribution.*

Honors & Awards

2024	**CIFRE PhD Scholar**, ANRT Association Nationale de la Recherche et de la Technologie	
2022	**Distinction**, For coursework and thesis, University of Oxford	
2020	**Academic Excellence Award**, For all years 2017, 2018, 2019, 2020 at IIT Kanpur	
2017	**All India Rank 168**, Joint Entrance Exam Advanced, 200,000 candidates	
2016	**Indian Olympiad Teams**, Selected for Indian teams in Mathematics (RMO) and Physics (NSEP)	

Talks & Teaching

2026	**Quantum Machine Learning: from Fundamentals to Applications**, Nordita Workshop, Invited Speaker — *invited tutorial-style lecture series at international QML workshop*	*Stockholm*
2025	**Bayesian Quantum Orthogonal Neural Networks for Anomaly Detection**, QTML 2025, Long Oral — *competitive oral selection at premier quantum ML conference*	*Singapore*
2025	**Instructor**, ACM India Summer School on Quantum Circuits and Quantum Algorithms — *designed and delivered lectures on variational quantum algorithms*	*JUIT, Solan, India*
2024	**Training-efficient density quantum machine learning**, QTML 2024, Long Oral	*Melbourne*
2021	**Teaching Assistant**, Data Structures and Algorithms, under Prof. Anil Seth	*IIT Kanpur*
2020	**Project Mentor**, ProML, Association of Computing Activities	*IIT Kanpur*

Skills

Languages	Python, C/C++
Quantum	Qiskit, PennyLane, Cirq
ML/Scientific	JAX, PyTorch, NumPy, SciPy
NLP/LLM	HuggingFace, BERT, GPT-2, Scikit-learn
Infrastructure	Git, AWS, LaTeX